Final Cybersecurity Assessment Report

Skyline Innovations, a fast-growing provider of cloud-based business solutions, has recently experienced multiple cybersecurity incidents that threaten corporate data, customer trust, and financial stability. A malicious mobile application was discovered on corporate devices, phishing attacks compromised employee credentials, and cloud misconfigurations exposed sensitive data.

This report is intended for the Chief Information Security Officer (CISO), Chief Technology Officer (CTO), and Executive Leadership of Skyline Innovations. It provides a detailed analysis of the identified security risks, an assessment of their potential impact, and actionable recommendations to enhance the company's cybersecurity posture.

Cybersecurity threats continue to evolve, and companies handling sensitive customer data must proactively address security weaknesses before they lead to data breaches or regulatory penalties. This report serves as a roadmap to mitigate current risks and establish long-term security resilience for Skyline Innovations.

To: Chief Information Security Officer, Chief Technology Officer, Executive Leadership

From: Tyler Michel, Cybersecurity Specialist

Date: April 19, 2026

Subject: Cybersecurity Assessment and Hardening Recommendations

1. Executive Summary

1. A sophisticated spyware Android application disguised as "Google Service 12.0" was installed on corporate devices.
2. Targeted phishing emails impersonating IT support have already resulted in credential compromises.
3. Misconfigured AWS cloud infrastructure, including overly permissive IAM roles, absent logging/monitoring, and publicly accessible S3 buckets.

These vulnerabilities allow for data exfiltration, account takeovers, intellectual property theft, and regulatory non-compliance. Some hardening recommendations include mobile device management enforcement, mandatory security awareness training with phishing simulations, and full AWS hardening. Implementing these measures aligned with the NIST Cybersecurity Framework, ISO 27001, and SANS policy.

2. Summary of Findings

Mobile Threat Analysis

A malicious Android application with specs of:

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1. **name:** com.my.spy.app
2. **file hash:** (5809066a109718683fa1ffe3abcd0e6c9bd5f613279e081e31bc17e628d9bfba.apk)

The file was discovered on corporate devices, and the app is disguised as a legitimate "Google service 12.0)" but is purpose-built spyware.

Mobile Security Framework analysis produced a security score of **53/100** with the following key findings:

1. **Dangerous and over-privileged permissions**:
    1. ACCESS_FINE_LOCATION
    2. ACCESS_BACKGROUND_LOCATION
    3. READ_SMS/RECEIVE_SMS/SEND_SMS
    4. PROCESS_OUTGOING_CALLS, READ_PHONE_STATE
    5. CALL_PHONE, RECORD_AUDIO, CAMERA
    6. READ_EXTERNAL_STORAGE/WRITE_EXTERNAL_STORAGE
    7. SYSTEM_ALERT_WINDOW, and READ_CONTACTS.
2. **Manifest and component exposure**:
    1. Android:allowBackup=true
    2. Multiple exported activities/services/receivers with insufficient protection.
    3. High intent priority.
3. **Code and anti-analysis techniques**:
    1. Anti-VM detection
    2. R8 compiler without markers
    3. Insecure random number generator
    4. Possible hardcoded secrets, and IP address disclosure.
4. **Certificate Issues**: Uses the default Android debug/test certificate, which is vulnerable to Janus attack and SHA1 collisions.

Risks: Silent exfiltration of location data, SMS, call logs, contacts, audio recordings, photos, and corporate credentials.

Phishing Attacks

Phishing email

Subject: URGENT: Mandatory Password Reset Required – Action Needed by 12:00 PM Today

From: Skyline IT Support <it-support@skyline-innovations-security.com>

Dear Skyline Team,

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We have detected suspicious login attempts on several employee accounts. As a security precaution, all users are required to reset their passwords immediately. Click the link below to verify your account and complete the secure password reset:

RESET PASSWORD NOW

This is a mandatory security update. Failure to reset your password by 4:00 PM today will result in a temporary account lockout.

Thank you for your immediate attention. Skyline Innovations IT Security Team

Note: This link is safe and expires in 24 hours.

Why employees fell for it: The message mimics official IT Staff, creates a false sense of security and exploits authority bias and urgency.

Impact: Compromised credentials enable account takeovers and lateral/vertical movement.

Cloud Security Vulnerabilities

Skyline's AWS infrastructure contained multiple critical misconfigurations:

1. **Excessive IAM role permissions:** Violates the principle of least privilege.
2. **Lack of logging and monitoring**: No AWS CloudTrail or CloudWatch alerts for suspicious activity.
3. **Exposed S3 buckets**: Public access not blocked, no encryption by default.

3. Recommendations

Mobile Security Improvements

1. Deploy an enterprise Mobile Device Management/Enterprise Mobility Management solution to enforce app allow-listing, block sideloading, and remotely quarantine suspicious apps.
2. Enable Google Play Protect and Mobile Threat Defense tools on all corporate Android devices.
3. Implement a formal Mobile Device Security Policy.
4. Conduct regular MobSF-style scans for all corporate devices.

Phishing Mitigation Strategies

1. Mandate security awareness training with quarterly simulated phishing campaigns and immediate corrective training for those who fail.
2. Deploy email security gateways with link sandboxing, domain impersonation detection, and DMARC/SPF/DKIM enforcement.
3. Require MFA on all accounts and adopt a zero-trust model.
4. Adopt SANS Acceptable Use Policy and Email Policy templates

5. Update the Employee Security Awareness Policy.

Cloud Security Hardening

1. Enable MFA on all IAM users and create users with least-privilege roles.
2. Implement a restrictive S3 bucket policy enforcing HTTPS only and block all public access.
3. Enable AWS CloudTrail with log validation and CloudWatch alarms for suspicious/unauthorized IAM actions.
4. Apply default encryption and server access logging on all S3 buckets.

4. Conclusion

Not implementing the recommendations can lead to severe data breaches, financial losses, reputational damage, and loss of customer trust. Implementing the recommendations above will help Skyline protect sensitive customer data and reduce the company's attack surface.

Proposed Timeline:

1. Short-term:
    a. Complete AWS hardening and launch phishing training.
    b. Quarantine mobile malware and force password resets for affected accounts
2. Long-term: Establish continuous monitoring, policy updates, and quarterly assessments.

Executive leadership is strongly encouraged to allocate necessary resources for implementing these security measures.

5. Supporting Evidence

1. **Mobile security scan results:**

APP SCORES

Security Score 53/100
Trackers Detection 1/432

FILE INFORMATION

File Name
5809066a109718683fa1ffe3abcd0e6c9bd5f613279e081e31bc17e628d9bfba.apk
Size 11.78MB
MD5 dd28810e5eb0aa85c71fd5e1211b7a2a
SHA1 b493cc7f024461f771be2b5fc8e24befaf3fa0ac
SHA256 5809066a109718683fa1ffe3abcd0e6c9bd5f613279e081e31bc17e628d9bfba

APP INFORMATION

App Name Google service
Package Name com.my.spy.app
Main Activity
com.my.spy.app.ui.activities.login.LoginActivity
Target SDK 33 Min SDK 21 Max SDK
Android Version Name 12.0 Android Version Code 12

FIREBASE DATABASES ANALYSIS

TITLE	SEVERITY	DESCRIPTION
App talks to a Firebase database	info	The app talks to Firebase database at https://my-spy-a9c92.firebaseio.com
Firebase Remote Config disabled	secure	Firebase Remote Config is disabled for https://firebaseremoteconfig.googleapis.com/v1/projects/1050903830580/namespaces/firebase:fetch?key=AIzaSyDQaFP4U8TzQToxPbVHg-h8cCOsZKwsdBA. This is indicated by the response: {'state': 'NO_TEMPLATE'}

ABUSED PERMISSIONS

TYPE	MATCHES	PERMISSIONS
Malware Permissions	16/25	android.permission.ACCESS_FINE_LOCATION, android.permission.WRITE_EXTERNAL_STORAGE, android.permission.READ_EXTERNAL_STORAGE, android.permission.READ_PHONE_STATE, android.permission.RECORD_AUDIO, android.permission.READ_CONTACTS, android.permission.RECEIVE_SMS, android.permission.READ_SMS, android.permission.SEND_SMS, android.permission.CAMERA, android.permission.SYSTEM_ALERT_WINDOW, android.permission.ACCESS_NETWORK_STATE, android.permission.INTERNET, android.permission.RECEIVE_BOOT_COMPLETED, android.permission.VIBRATE, android.permission.WAKE_LOCK
Other Common Permissions	8/44	android.permission.ACCESS_BACKGROUND_LOCATION, android.permission.PROCESS_OUTGOING_CALLS, android.permission.CALL_PHONE, android.permission.WRITE_SMS, android.permission.REQUEST_IGNORE_BATTERY_OPTIMIZATIONS, android.permission.PACKAGE_USAGE_STATS, com.google.android.gms.permission.AD_ID, com.google.android.finsky.permission.BIND_GET_INSTALL_REFERRER_SERVICE

Malware Permissions:
Top permissions that are widely abused by known malware.
Other Common Permissions:
Permissions that are commonly abused by known malware.

OFAC SANCTIONED COUNTRIES

This app may communicate with the following OFAC sanctioned list of countries.

DOMAIN	COUNTRY/REGION

DOMAIN MALWARE CHECK

DOMAIN	STATUS	GEOLOCATION

2. **The phishing email used in the attack:**

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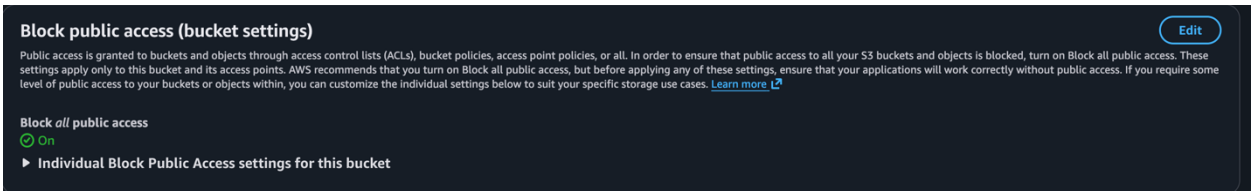
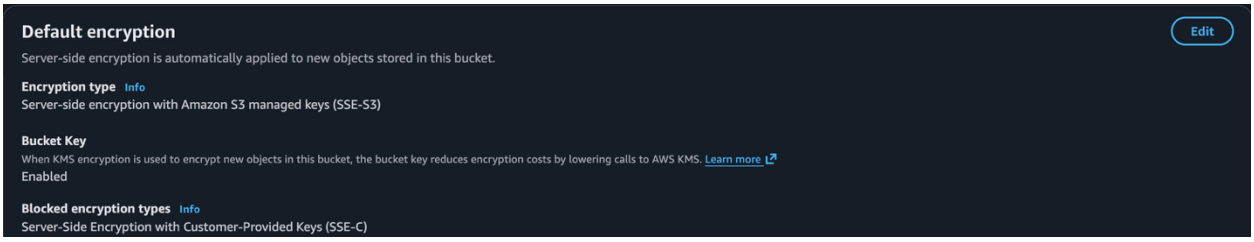
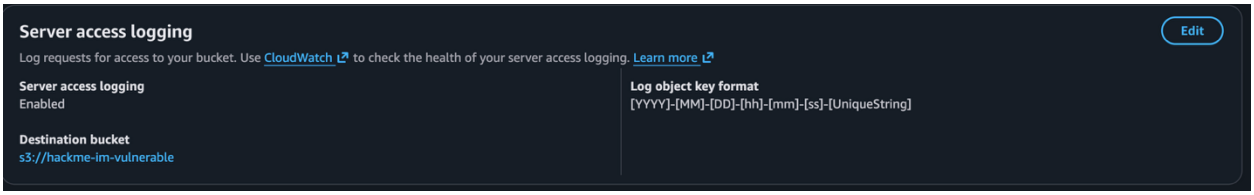
3. AWS security configurations before and after hardening:

Only HTTPS is allowed



Encryption is enabled



CloudWatch is enabled and running